Statistical Methods
Exam Solution - Moed A

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Question A

See Hebrew question in Word doc

Solution

1. (a) The likelihood function reads:

$$L(y;\theta) = \prod_{i=1}^{n} \frac{\theta^y}{y!} e^{-\theta}$$

$$logL = \sum_{i=1}^{n} [y_i log(\theta) - \theta - log(y!)]$$

The maximum of this function will be given by vanishing the derivative:

$$0 = \frac{\partial logL}{\partial \theta} = \sum_{i=0}^{n} \frac{y_i}{\theta} - 1$$

$$\hat{\theta} = \frac{1}{n} \sum_{i=0}^{n} y_i$$

(b)

$$E[\hat{\theta}] = \frac{1}{n} \sum_{i=0} nE[y_i] = \frac{1}{n} \cdot n \cdot \theta = \theta$$

$$Var(\hat{\theta}) = \frac{1}{n^2} \sum_{i=0}^{n} Var(y_i) = \frac{1}{n^2} \cdot \theta = \frac{\theta}{n}$$

$$Bias(\hat{\theta}) = E[\hat{\theta}] - \theta = \theta - \theta = 0 \rightarrow \text{Unbiased}$$

2. (a) The mean:

$$E[\hat{\theta}_1] = \frac{E[Y_1] + E[Y_2]/2}{2} = \frac{\theta + 2\theta/2}{2} = \theta$$

$$Bias(\hat{\theta}_1) = 0 \rightarrow \text{ Unbiased}$$

By using $var(aX) = a^2 var(X)$ the variance reads:

$$Var[\hat{\theta}_1] = \frac{Var[Y_1] + Var[Y_2]/4}{4} = \frac{1}{4}var(Y_1) + \frac{1}{16}var(Y_2) = \frac{1}{4}\sigma_1^2 + \frac{1}{16}\sigma_2^2$$

The variance is also the MSE.

(b) **The likelihood**

$$L(\theta; e_1, e_2) = \frac{1}{\sqrt{2\pi}} exp\left(-\frac{e_1^2}{2}\right) \frac{1}{\sqrt{2\pi}} exp\left(-\frac{e_2^2}{2}\right) = \frac{1}{2\pi} exp\left(-\frac{e_1^2 + e^2}{2}\right)$$

Log-Likelihood σ^2

$$LL(\theta) = -log(2\pi) - \frac{1}{2}\left((Y_1 - \theta)^2 + (Y_2 - 2\theta)^2\right)$$

Partial derivative with respect to θ

$$\frac{\partial}{\partial\theta} LL = (Y_1 - \theta) + 2(Y_2 - 2\theta) = 0 \Rightarrow \hat{\theta}_2 = \frac{Y_1 + 2Y_2}{5}$$

The ML estimator is also unbiased:

$$E[\hat{\theta}_2] = \frac{E[Y_1] + 2E[Y_2]}{5} = \theta$$

The variance:

$$Var[\hat{\theta}_2] = \frac{Var[Y_1] + 2^2 Var[Y_2]}{5^2} = \frac{1}{25}\sigma_1^2 + \frac{4}{25}\sigma_2^2$$

The variance is also the MSE.

(c) For $\sigma_1 = \sigma_2 = 1$ we get: $Var(\hat{\theta}_1) = \frac{5}{16}$ and $Var(\hat{\theta}_2) = \frac{5}{25}$
Since both estimators are unbiased and $\hat{\theta}_2$ has smaller variance, we will choose $\hat{\theta}_2$.

0.1 Question B

Let $W = \begin{bmatrix} W_1 \\ W_2 \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right)$

And let A be a random variable, statistically independent of W.

We define $X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = A \begin{bmatrix} W_1 \\ W_2 \end{bmatrix}$

1. Find the conditional PDF $f_{X|A}(X|A = a)$
2. In this section, assume $A \sim N(0, \sigma^2)$,
   (a) Find the optimal MMSE estimators for X_1^2 and X_2^2 given A.
   (b) Calculate the Mean Square Error (MSE) of the two estimators.
3. In this section, assume $A \sim Bernoulli(0.5)$, namely, $A = \begin{cases} 0 & \text{with probability } 0.5 \\ 1 & \text{with probability } 0.5 \end{cases}$

   We define a new random variable $Y = AW_1 + Z$ where $Z = N(0, 1)$ statistically independent of A and of W_1. Find the optimal MMSE estimator for W_1 given Y.

Solution

1. We are interested in the conditional PDF $f_{X|A}(X|A = a)$. But, given $A = a$ ... n of the Gaussia ... calculate its expe ...

   C_X ... $a^2 I$

   To conc ...

2. (a) In ... and now we ...

   (b) ...

   M ... $+E[X_1^4]$

   We ... formula for ...

   $= E[A^4] - 2E_A[A^4] + E_A[3A^4] = 2E[A^4] = 6\sigma^4$

   The last equation used the given $A \sim N(0, \sigma^2)$.

Wrong solution – see pages below

3.

$$\hat{W}_1 = E[W_1|Y] = E[E[W_1|Y, A]|Y]$$

Given A: Y and W_1 are jointly Gaussian, and therefore $E[W_1|Y, A]$ is equal to the BLE of W_1 given Y (conditioned on A):

$E[W_1|A]$

$E[Y^2|A]$ … $W_1 Z] =$

$A\ E[W_1$

Plug to

$= Y$ … $|Y)$

Wrong solution – see pages below

For the

$Pr(A =$ … $= y|A = 1)$ $=$

Given A

Given $A = 1$ we have $Y|A = W + Z \sim N(0, 2)$

$$f(Y = y|A = 1) = \frac{1}{\sqrt{2\pi \cdot 2}} exp(-\frac{y^2}{2 \cdot 2})$$

Hence,

$$Pr(A = 1|Y = y) = \frac{1}{1 + \sqrt{2} exp(-\frac{y^2}{4})}$$

שאלה 2 (מועד א 2020)

נתון

$$W=\left(\begin{array}{c}W_1\\W_2\end{array}\right)\sim N\left(\left(\begin{array}{c}0\\0\end{array}\right),\left(\begin{array}{cc}1&0\\0&1\end{array}\right)\right)$$

ונתון A משתנה אקראי בלתי תלוי סטטיסטית בW. נגדיר

$$X=\left(\begin{array}{c}X_1\\X_2\end{array}\right)=A\left(\begin{array}{c}W_1\\W_2\end{array}\right)$$

1. מצאו את הpdf של ההסתברות המותנת $f_{X|A}\left(X|A\right)$
2. בסעיף זה הניחו $A\sim N\left(0,\sigma^2\right)$
   (א) מצאו משערך $MMSE$ של X_1^2 בהינתן A.
   (ב) חשבו את הMSE של המשערך. הדרכה: השתמשו בכלל התוחלת השלמה. ובכך שאם $x\sim N\left(0,\sigma^2\right)$ אז $E\left[x^4\right]=3\sigma^2$
3. בסעיף זה הניחו שA מתפלג לפי:

$$p\left(A\right)=\begin{cases}0.5 & A=0\\0.5 & A=1\end{cases}$$

ונגדיר משתנה אקראי חדש $Y=AW_1+W_2$. מצאו משערך W_1 בהינתן Y.
הדרכה: חשבו את $E\left[W_1|A,Y\right]$ והשתמשו בלל התוחלת השלמה.

פתרון

1. $f_{X|A}\left(X|A\right)$ מתפלג נורמלית מכיוון שהוא כפל בסקלר של התפלגות נורמלית. נחשב את הפרטמרים של ההתפלגות:

$$E\left[X|A\right]=E\left[A\left(\begin{array}{c}0\\0\end{array}\right)\right]=\left(\begin{array}{c}0\\0\end{array}\right)$$
$$VAR\left(X|A\right)=VAR\left(A\left(\begin{array}{cc}1&0\\0&1\end{array}\right)\right)=\left(\begin{array}{cc}A^2&0\\0&A^2\end{array}\right)$$

כלומר

$$f_{X|A}\left(X|A\right)\sim N\left(\left(\begin{array}{c}0\\0\end{array}\right),\left(\begin{array}{cc}A^2&0\\0&A^2\end{array}\right)\right)$$

2. עבור A מ"מ גאוסי:
   (א) משערך ה$MMSE$ הוא:

$$\hat{X_1^2}=E\left[X_1^2|A\right]=VAR\left(X_1|A\right)+E^2\left[X_1|A\right]=A^2+0=A^2$$

(ב) הMSE נתון ע"י:

$$\begin{aligned} E\left[\left(E\left[X_1^2|A\right]-X_1^2\right)^2\right] &= E\left[\left(A^2-X_1^2\right)^2\right] = E\left[A^4-2X_1^2A^2+X_1^4\right] \\ &= E\left[E\left[\left(A^4-2X_1^2A^2+X_1^4\right)|A\right]\right] \\ &= E\left[A^4-2A^2A^2+3A^4\right] = 2E\left[A^4\right] = 6\sigma^2 \end{aligned}$$

3. נשתמש בכלל התוחלת השלמה:

$$\hat{W}_1 = E\left[W_1|Y\right] = E\left[E\left[W_1|A,Y\right]|Y\right] = \sum_{A=0,1} p\left(A|Y\right)E\left[W_1|A,Y\right]$$

כעת נשים לב ש$\begin{pmatrix} W_1 \\ Y \end{pmatrix}|A$ הוא וקטור גאוסי (כמו שראינו בתרגול), כאשר:

$$\begin{pmatrix} \mu_{w_1} \\ \mu_Y \end{pmatrix}|A\Bigg) = E\left[\begin{pmatrix} W_1 \\ Y \end{pmatrix}|A\right] = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} C_{W_1W_1} & C_{W_1Y} \\ C_{W_1Y} & C_{YY} \end{pmatrix} = COV\left(\begin{pmatrix} W_1 \\ Y \end{pmatrix}|A\right) = \begin{pmatrix} 1 & A \\ A & A^2+1 \end{pmatrix}$$

לכן

$$E\left[W_1|A,Y\right] = \mu_{W_1} + C_{W_1Y}C_{YY}^{-1}\left(Y-\mu_Y\right) = \frac{A}{A^2+1}Y$$

כלומר

$$\begin{aligned} E\left[W_1|A=0,Y\right] &= 0 \\ E\left[W_1|A=1,Y\right] &= \frac{1}{2}Y \end{aligned}$$

כעת נחשב את $p\left(A|Y\right)$ באמצעות חוק בייס

$$p\left(A|Y\right) = \frac{p\left(Y|A\right)P\left(A\right)}{P\left(Y\right)} = \frac{p\left(Y|A\right)P\left(A\right)}{\sum_A p\left(Y|A'\right)P\left(A'\right)} =$$

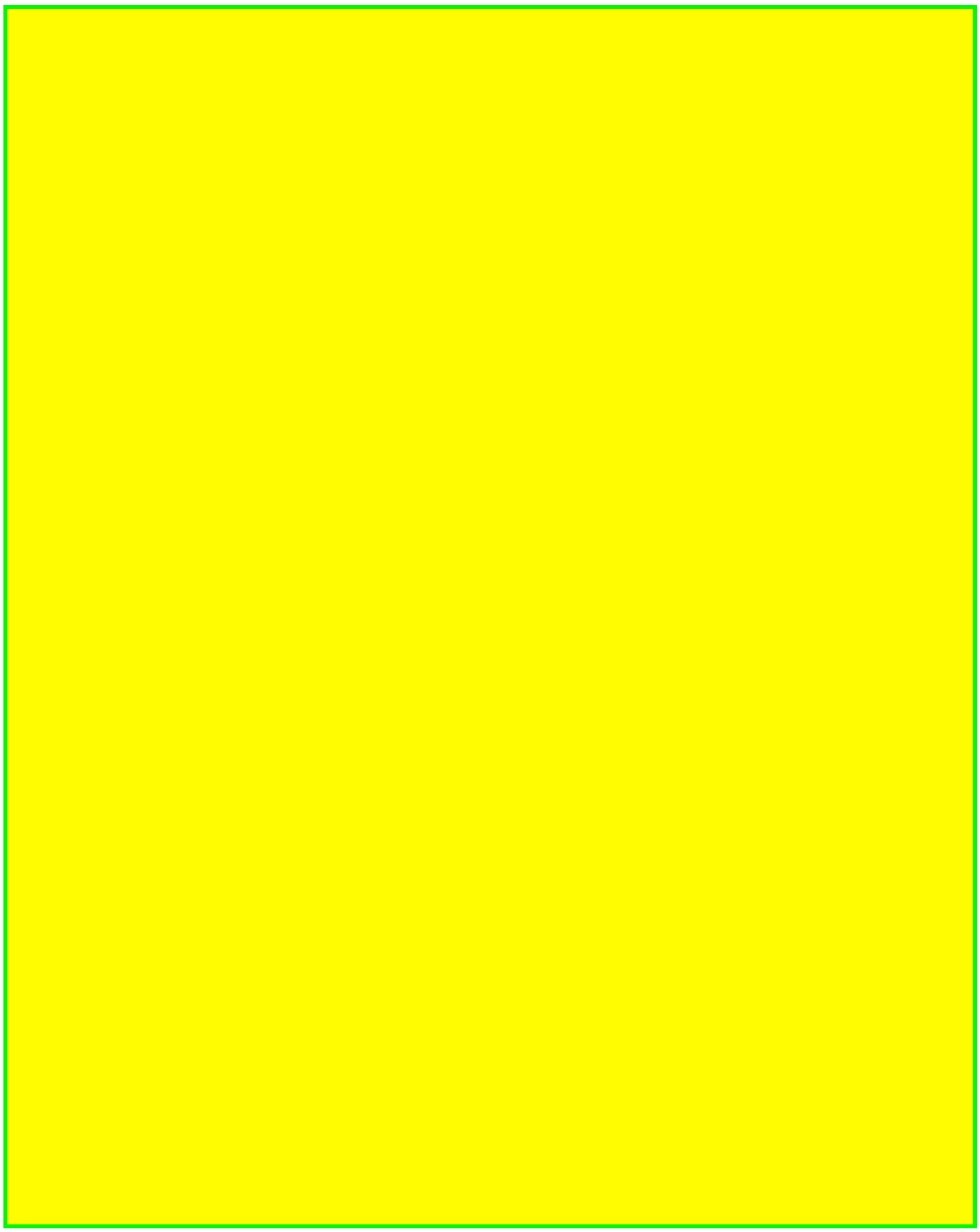
ומכיוון ש$Y|A \sim N\left(0,A^2+1\right)$

$$p\left(A|Y\right) = \frac{\frac{1}{\sqrt{2\pi(A^2+1)}}e^{-\frac{Y^2}{A^2+1}}\frac{1}{2}}{\sum_{A'}\frac{1}{\sqrt{2\pi(A'^2+1)}}e^{-\frac{Y^2}{A'^2+1}}\frac{1}{2}} = \frac{\frac{1}{\sqrt{(A^2+1)}}e^{-\frac{Y^2}{A^2+1}}}{\sum_{A'}\frac{1}{\sqrt{(A'^2+1)}}e^{-\frac{Y^2}{A'^2+1}}}$$

$$\begin{aligned} P\left(A=0|Y\right) &= \frac{e^{-\frac{Y^2}{2}}}{e^{-\frac{Y^2}{2}}+\frac{1}{\sqrt{2}}e^{-\frac{Y^2}{4}}} = \frac{1}{1+\frac{1}{\sqrt{2}}e^{\frac{Y^2}{4}}} \\ P\left(A=1|Y\right) &= \frac{\frac{1}{\sqrt{2}}e^{-\frac{Y^2}{4}}}{e^{-\frac{Y^2}{2}}+\frac{1}{\sqrt{2}}e^{-\frac{Y^2}{4}}} = \frac{1}{\sqrt{2}+e^{\frac{-Y^2}{4}}} \end{aligned}$$

לכן בסך הכל:

$$\begin{aligned}
\hat{W}_1 &= E\left[W_1 | Y\right] = \\
&= \frac{1}{1+\frac{1}{\sqrt{2}}e^{\frac{Y^2}{4}}}0 + \frac{1}{2}\frac{1}{\sqrt{2}+e^{\frac{-Y^2}{4}}}Y = \\
&= \frac{1}{2}\frac{1}{\sqrt{2}+e^{\frac{-Y^2}{4}}}Y
\end{aligned}$$



0.2 Question C

Consider :

$$\begin{aligned} W[n] &= {}_{\text{i.i.d}} \begin{cases} +1 & \text{w.p.} \quad 0.5 \\ -1 & \text{w.p.} \quad 0.5 \end{cases} \\ X[0] &= 0 \\ X[n] &= X[n-1] + W[n], \quad n = 1, \cdots \end{aligned}$$

Below, assume $n \geq 0$

1. What is the mean of $X[n]$?
2. What is the autocorrelation function of $X[n]$?
3. Is $X[n]$ a WSS process?
4. What is the BLE of $X[n]$ given $\{X[n-1], X[n-2], X[n-3]\}$?
5. What is the MMSE estimator of $X[n+2]$ given $X[n]$?

Solution

1. $E[X_n] = E[X_{n-1} + W_n] = E[X_{n-1}] = ... = E[X_0] = 0$
2. First note that the process can be also expressed as following:

$$X_n = X_{n-1} + W_n = X_{n-2} + W_{n-1} + W_n = ... = \sum_{i=0}^{n} W_i$$

   To calculate the autocorrelation (without loss of generality, for $k \geq 0$):

$$E[X_n X_{n+k}] = E\left[\left(\sum_{i=0}^{n} W_i\right)\left(\sum_{j=0}^{n+k} W_j\right)\right] = \sum_{i=0}^{n}\sum_{j=0}^{n+k} E[W_i W_j]$$

   For $i \neq j$ we get $E[W_i W_j] = E[W_i]E[W_j] = 0$ and therefore:

$$E[X_n X_{n+k}] = \sum_{i=0}^{n} E[W_i^2] = n$$

3. Immediate conclusion from the previous section is that the process is not WSS (e.g. the variance is not stationary $R(n, n) = n$.
4. Given X_{n-1} and W_n we get X_n deterministically. So the BLE given $\{X_{n-1}, X_{n-2}, X_{n-3}\}$ is the BLE given X_{n-1}. We will use the orthogonality principle to prove that the BLE estimator for X_n given X_{n-1} is simply $\hat{X_n} = X_{n-1}$. Let a, b be arbitrary constants and denote $g(x_{n-1}) = aX_{n-1} + b$. We need to show that $E[g(x_{n-1})(\hat{x_n} - x_n)] = 0$ for every a, b:

$$E[(ax_{n-1} + b)(x_n - x_{n-1})] = E[(ax_{n-1} + b)W_n] = 0$$

   We used the fact that W_n is independent of x_{n-1} and the linearity of the expectation.

5. To calculate the MMSE estimator for X_{n+2}^2 given X_n first we can write explicitly: $X_{n+2} = X_{n+1} + W_{n+2} = X_n + W_{n+1} + W_{n+2}$. And now we can calculate MMSE directly:

$$\hat{X}_{n+2}^2 = E[X_{n+2}^2 | X_n] = E[(X_n + W_{n+1} + W_{n+2})^2 | X_n] =$$

The three RVs are statistically independent with zero-mean, so expected value of multiplications are zero except the squares:

$$= E[X_n^2 | X_n] + E[W_{n+1}^2 | X_n] + E[W_{n+2}^2 | X_n] = X_n^2 + 2$$